

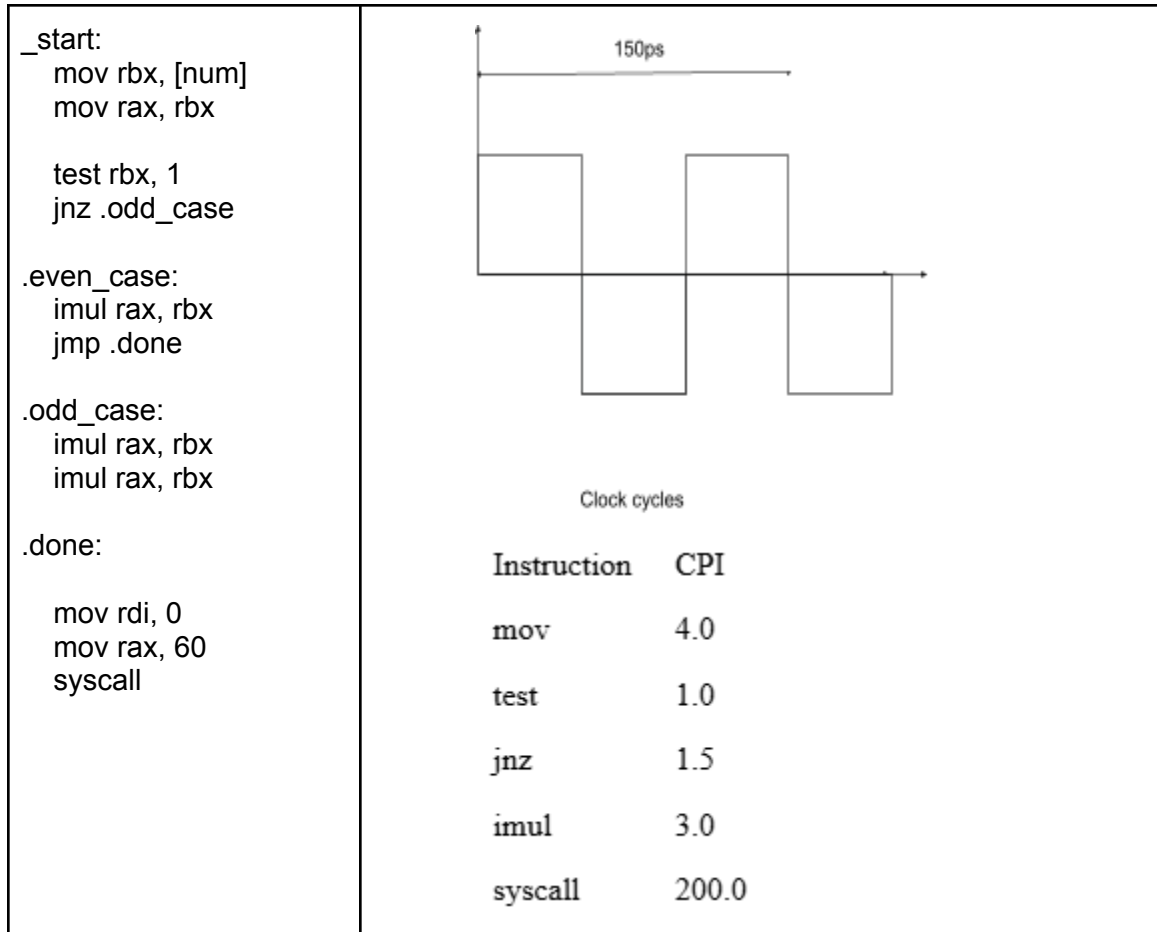
1. Can you relate/connect Amdahl's law with any of the design principles, mentioned below? Explain the reason with an example. [3]

	Design Principle
1	Performance via Prediction
2	Performance via Pipelining
3	Make the common case faster
4	Use Abstraction to Simplify Design

2. Define the term throughput and response time. How do these two differ, and why might one be prioritized over the other in different computing environments? [1+1+2]
3. A processor has the following instruction mix:
 - a. 50% ALU instructions with CPI = 1
 - b. 30% load/store with CPI = 2
 - c. 20% branch with CPI = 3
 Calculate the average CPI of the program. [3]
4. A computer runs a program in 20 seconds. A compiler optimization reduces instruction count by 15% and CPI by 10%, but the clock rate stays the same. What is the new execution time? [3]
5. To calculate the benchmark of a system, why do we take the geometric mean instead of only taking the average of the individual spec ratios? [3]
6. A researcher is benchmarking the floating-point performance of an Intel Core i7 15th Gen processor using a subset of SPECfp benchmarks. They record the reference execution times (from the official SPEC baseline machine) and the measured execution times (on the i7-15th Gen system). Calculate the final SPEC value, show all the necessary steps. [6]

Benchmark	Reference Time (s)	Measured Time (s)
470.lbm	2550	225
444.namd	2950	210
453.povray	3120	280
459.GemsFDTD	4200	400
482.sphinx3	3750	350

7. Mention if the following statement is true or false and also explain why.
A 4-core processor can always achieve a 4× speedup over a single-core version on any program. [1+1]
8. Consider the following assembly code snippet for Intel Core-i7 15th Gen(x86).



- Find the number of instructions in the above code. [1]
 - Find the average CPI. [2]
 - Find the clock cycle time. [1]
 - Find the time to execute this program. [2]
9. A CPU has a MIPS rating of 6. A program consists of: 10 million instructions, of which 60% are arithmetic (CPI = 2), 30% are memory access (CPI = 3), and 10% are control (CPI = 1). Determine:
- Average CPI [1]
 - Clock frequency of the processor [2]
 - Execution time of the program [2]